

Esame superato :
Nodi e Archi
corretti

Usare
trattino basso
attributi privati

Intro to Graphs

NetworkX

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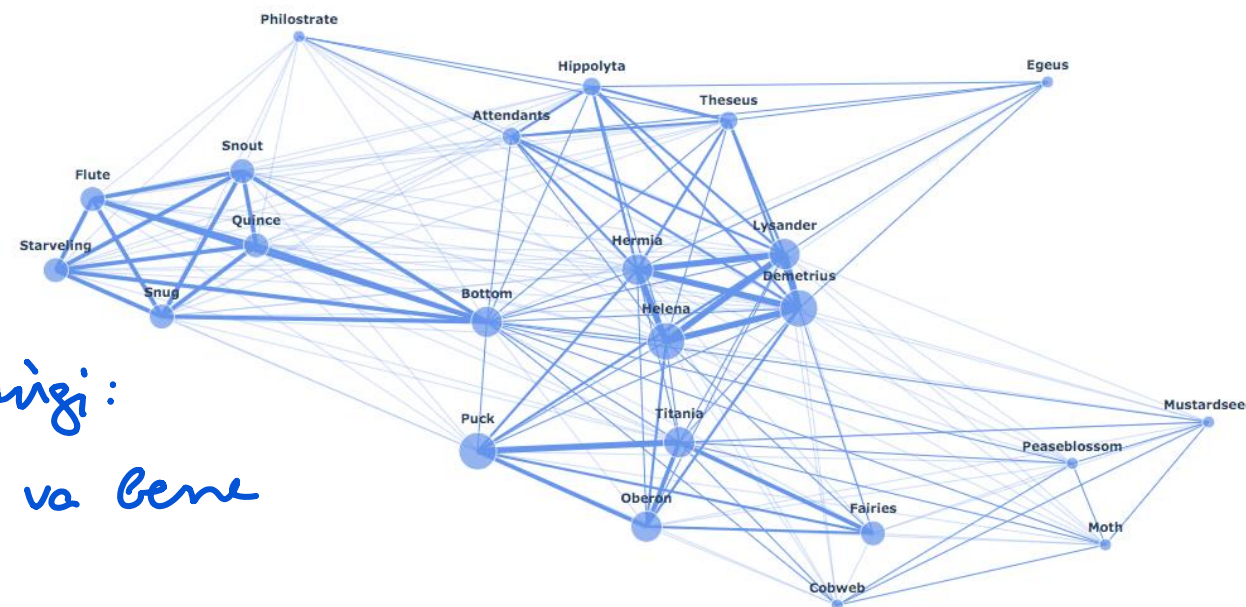
Esercizio metro-Paigi:

Se pochi nodi (≈ 10) va bene
MOD0 1

! archi uguali ne
viene salvato solo 1 !
ne non si usa
multigraph

! Se creo archi di
nodi non presenti,
crea anche i nodi !

fare controlli
critici



NODI → Dataclass

GRAFO → costruito con
networkx

`nx.DiGraph()`

1

+ di 1

`g.add_edge`

`g.add_node`

`g.add_edges_from`

`g.add_nodes_from`

3 nodi non
hanno un tipo
obbligatorio = a tutte

es) `modi = { 1, 2, "a", "b" }`

PESATO:

`g.add_weighted_edges_from`

`g.get_edge_data`

INTRODUCTION TO NETWORKX

Strutturare sempre il grafo all'inizio
della funzione di riempimento

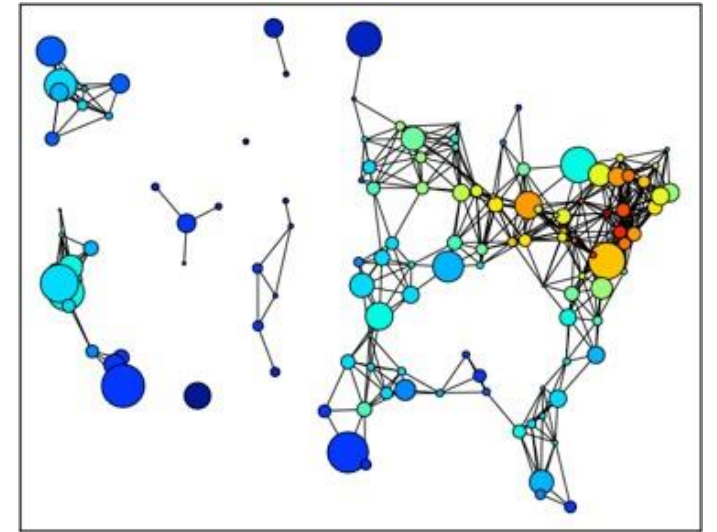
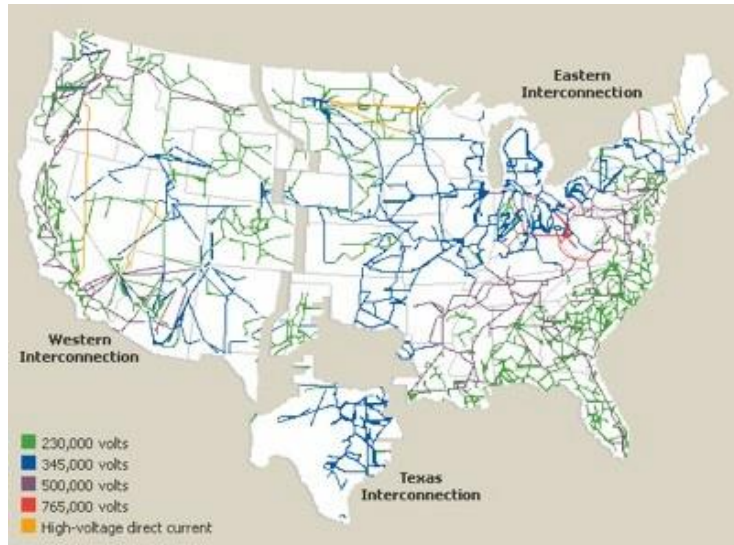
`g[nodo]` → nodi adiacenti,
con relativo
peso
`g[nodo1][nodo2]` → info sull'arco

Introduction to NetworkX - network analysis

- Vast amounts of network data are being generated and collected
- Sociology: web pages, mobile phones, social networks
- Technology: Internet routers, vehicular flows, power grids

How can we analyse these networks?

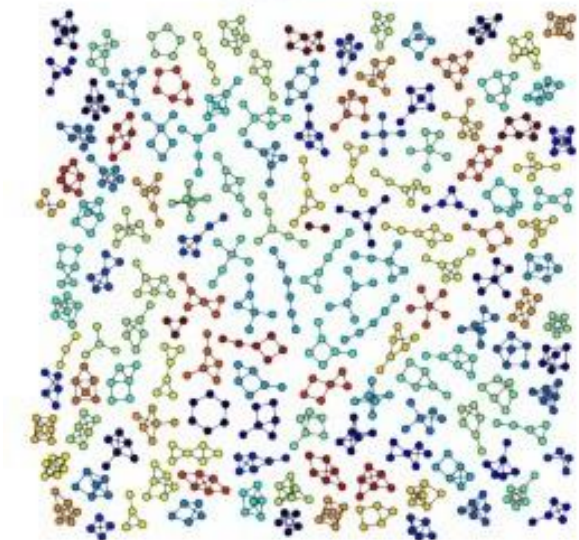
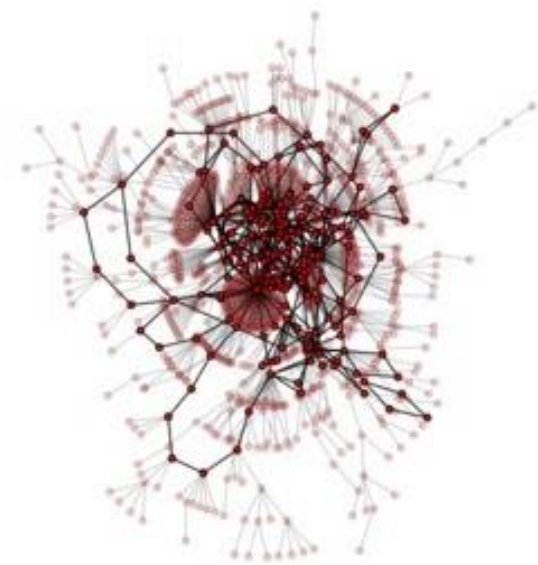
Python + NetworkX!



Introduction to NetworkX

“Python package for the creation, manipulation and study of the structure, dynamics and functions of complex networks.”

- Data structures for representing many types of data in the form of graphs
- Nodes can be any (hashable) Python object, edges can contain arbitrary data
- Flexibility ideal for representing networks found in many different fields
- Easy to install on multiple platforms
- Online up-to-date documentation
- First public release in April 2005



Introduction to NetworkX - design requirements

- Tool to study the structure and dynamics of social, biological, and infrastructure networks
- Ease-of-use and rapid development
- Open-source tool base that can easily grow in a multidisciplinary environment with non-expert users and developers
- An easy interface to existing code bases written in C, C++, and FORTRAN
- To painlessly slurp in relatively large nonstandard data sets

Introduction to NetworkX - object model

NetworkX defines no custom node objects or edge objects

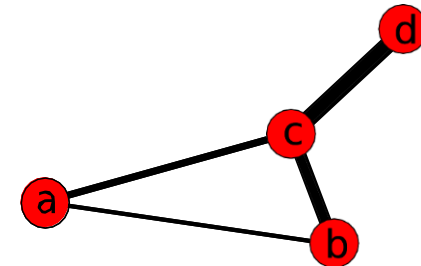
- node-centric view of network
- nodes can be any hashable object, while edges are tuples with optional edge data (stored in dictionary)
- any Python object is allowed as edge data and it is assigned and stored in a Python dictionary (default empty)

Introduction to NetworkX - quick example

- Search for the shortest path in a weighted and unweighted network:

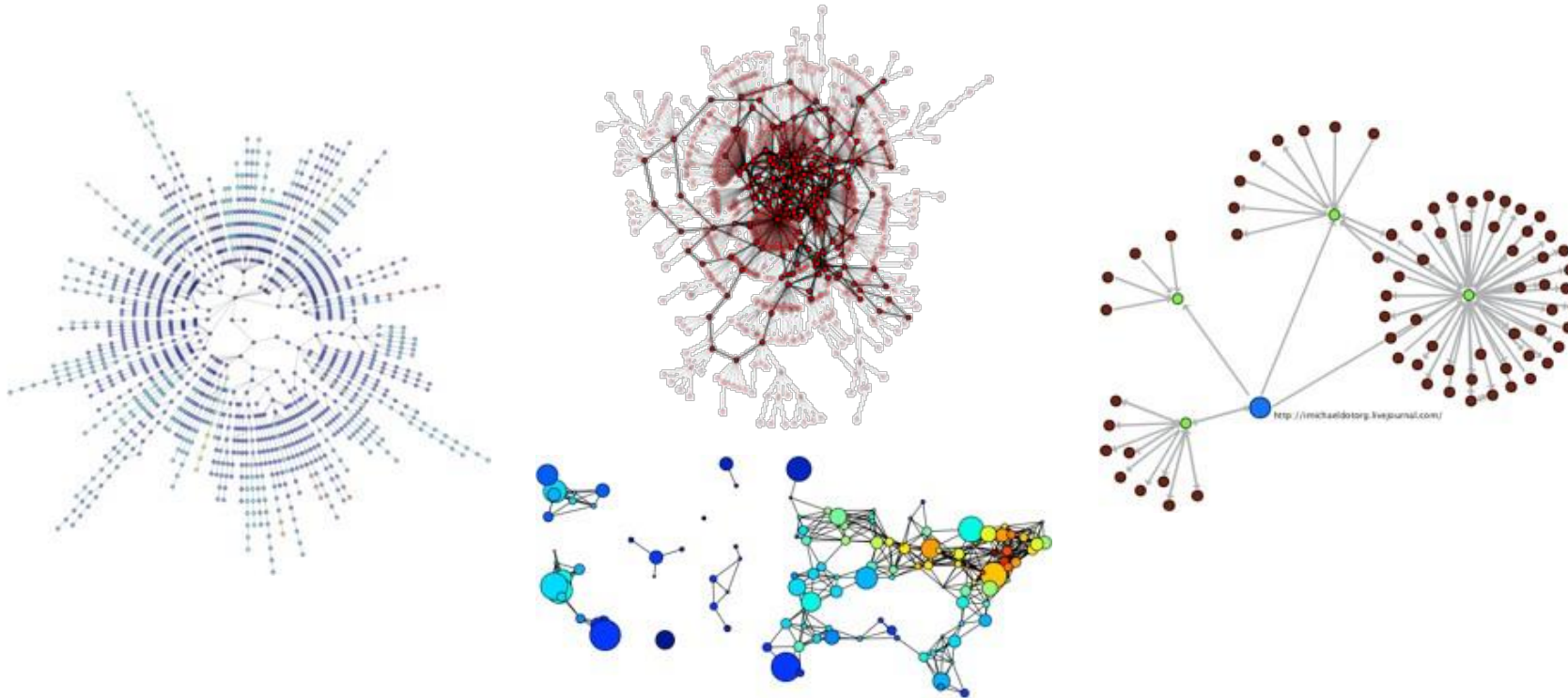
```
1 import networkx as nx
2 g = nx.Graph()
3 g.add_edge("a", "b", weight=1)
4 g.add_edge("b", "c", weight=100)
5 g.add_edge("a", "c", weight=1)
6 g.add_edge("c", "d", weight=1)
7 print(nx.shortest_path(g, "b", "d"))
8 print(nx.dijkstra_path(g, "b", "d", weight='weight'))
```

*Si possono anche
solo definire
gli archi.
Automaticamente
crea i nodi*



Introduction to NetworkX - drawing and plotting

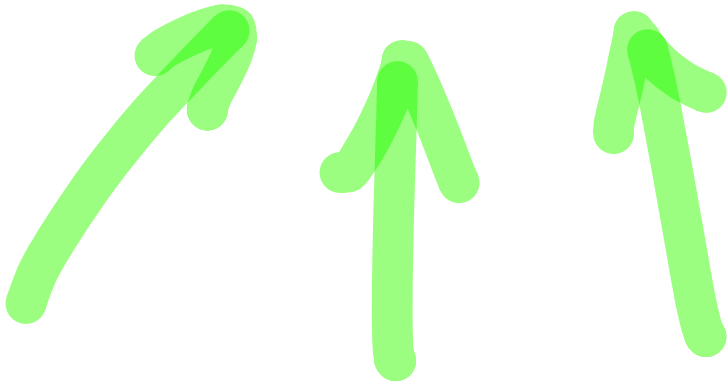
- It is possible to draw small graphs within NetworkX and to export network data and draw with other programs (i.e., GraphViz, matplotlib)



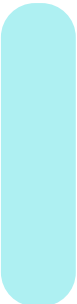
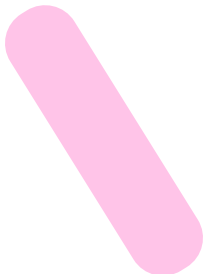
Introduction to NetworkX - official website

- <https://networkx.org/>

! Vedere se ci sono
• funzioni utili
all'esame !



A screenshot of the NetworkX official website. The header includes the NetworkX logo and navigation links: Install, Tutorial, Reference (highlighted), Gallery, Developer, Releases, and Guides. A search bar and a version dropdown (3.2.1 stable) are on the right. The left sidebar shows a 'Section Navigation' menu with links to Introduction, Graph types, Algorithms, Functions, Graph generators, Linear algebra, Converting to and from other data formats, Relabeling nodes, Reading and writing graphs, Drawing, Randomness, Exceptions, Utilities, and Glossary. The main content area is titled 'Reference' and shows the current release as 3.2.1, dated Oct 28, 2023. Below this, there are links for Introduction, NetworkX Basics, Graphs, Graph Creation, Graph Reporting, Algorithms, Drawing, Data Structure, Graph types, Which graph class should I use?, Basic graph types, Graph Views, Core Views, Filters, Algorithms, Approximations and Heuristics, and Assortativity.



GETTING STARTED WITH PYTHON AND NETWORKX

Getting started – import NetworkX

- NetworkX supports many different graph types, like:
 - `nx.Graph()` – undirected
 - `nx.DiGraph()` – directed
 - `nx.MultiGraph()` – supports multiple edges between nodes
 - `nx.MultiDiGraph()` – directed multigraph
- Also provide implementation of notable graphs (like heawood)

```
1 import networkx as nx
2 import math
3 import flet as ft
4
5 g = nx.heawood_graph()
6 print(g.nodes, g.edges)
7
8 g.add_node(math.cos) # cosine functionx
9 g.add_node(ft.Text("Pippo"))
10 g.add_edge("math.cos", 3)
11 print(g.nodes, g.edges)
```

Getting started – build a graph

- Nodes could be (almost) anything
 - Numbers, strings
 - Objects
 - Functions
 - Flet containers
- Edges connect nodes (even heterogeneous)
- Nodes and edges could have attributes

```
import networkx as nx
import math
import flet as ft

g = nx.Graph()
g.add_edge(1, 2) # default edge data=1
g.add_edge(2, 3, weight=0.9) # specify edge data

g.add_edge('y', 'x', function=math.cos)
g.add_node(math.cos) # any hashable can be a node

elist = [(1, 2), (2, 3), (1, 4), (4, 2)]
g.add_edges_from(elist)
elist = [('a', 'b', 5.0), ('b', 'c', 3.0), ('a', 'c', 1.0), ('c', 'd', 7.3)]
g.add_weighted_edges_from(elist) ←
g.add_node(ft.Text("Pippo"))

print(g.nodes())
print(g.edges())
print(g.get_edge_data('a', 'b')) ←
```

Getting started – Data Structure

- A graph is essentially a “dictionary of dictionaries of dictionaries”
- The keys are the nodes
- Indeed, `g[n]` yields a dictionary where keys are all the nodes connected with `n` (adjacency) and values are the edges params (like weight)

```
import networkx as nx

g = nx.Graph()
g.add_edge(1, 2) # default edge data=1
g.add_edge(2, 3, weight=0.9) # specify edge data

elist = [(1, 2, 1), (2, 3, 1), (1, 4, 1), (4, 2, 1),
         ('a', 'b', 5.0), ('b', 'c', 3.0), ('a', 'c', 1.0), ('c', 'd', 7.3)]
g.add_weighted_edges_from(elist)

print(g[2])
```

```
{1: {'weight': 1}, 3: {'weight': 1}, 4: {'weight': 1}}
```

```
Process finished with exit code 0
```

Getting started – Data Structure

- `g[u][v]` yields the edge attributes
- `n in g` tests if node `n` is in `g`
- `for n in g:` iterates through the graph
- `for nbr in g[n]:` iterates through the neighbors of `n`
- Data struct for directed graphs is only slightly more complex (two dicts, one for successors and one for predecessors)
- You can also use `g.nodes()` and `g.edges()` to get corresponding data
- Edges can have arbitrary attributes

```
import networkx as nx

g = nx.Graph()
g.add_edge(1, 2) # default edge data=1
g.add_edge(2, 3, weight=0.9) # specify edge data

elist = [(1, 2, 1), (2, 3, 1), (1, 4, 1), (4, 2, 1),
          ('a', 'b', 5.0), ('b', 'c', 3.0), ('a', 'c', 1.0), ('c', 'd', 7.3)]
g.add_weighted_edges_from(elist)
g.add_edge(2, 5, arbitraryAttr = "foo")

print(g[2])
print("-----")
print(g['a']['b'])
print("-----")
print('e' in g)
print("-----")
for n in g:
    print(n)
print("-----")
for nbr in g[2]:
    print(nbr)
print("-----")
print(g[2][5]['arbitraryAttr'])
```


Getting started – Directed and Multi

- Graphs can be directed, therefore differentiating neighbors in predecessors and successors
- Two nodes can have more than one edge

```
import networkx as nx

dg = nx.DiGraph()
dg.add_weighted_edges_from([(1,4,0.5), (3,1,0.75)])

print([s for s in dg.successors(1)])
print([p for p in dg.predecessors(1)])

mg = nx.MultiGraph()

mg.add_weighted_edges_from([(1,2,.5), (1,2,.75),
(2,3,.5)])

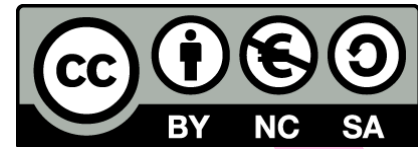
print(mg[1][2])
```

Getting started - graph operators

Classic graph operations

- `subgraph(G, nbunch)` - induce subgraph of G on nodes in nbunch
- `union(G1, G2)` - graph union
- `disjoint_union(G1, G2)` - graph union assuming all nodes are different
- `compose(G1, G2)` - combine graphs identifying nodes common to both
- `complement(G)` - graph complement
- `create_empty_copy(G)` - return an empty copy of the same graph class
- `convert_to_undirected(G)` - return an undirected representation of G
- `convert_to_directed(G)` - return a directed representation of G

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